

STRUISBAAI, South Africa

WMO: 689210

Lat: 34.8003S

Lon: 20.0569E

Elev: 3

StdP: 101.29

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.5	6.7	3.1	4.7	7.9	3.9	5.0	9.0	12.8	16.6	12.0	15.8	2.3	290	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.4	25.2	20.7	24.6	20.3	23.9	19.9	22.1	24.0	21.5	23.5	21.0	23.1	5.7	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	16.1	23.4	20.7	15.4	23.0	20.1	14.8	22.6	64.8	24.1	62.7	23.6	60.6	23.1	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.3	11.1	9.9	DB	2.6	27.9	1.6	1.5	1.5	28.9	0.6	29.8	-0.3	30.6	-1.4	31.7
			WB	2.1	22.5	0.9	1.1	1.5	23.2	0.9	23.9	0.4	24.5	-0.3	25.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.9	21.2	21.3	20.0	17.5	15.4	13.2	12.7	13.1	14.6	16.4	18.0	20.2	(6)
	DBStd	3.57	1.67	1.46	1.71	1.89	1.99	1.87	2.04	1.90	1.90	1.84	1.63	1.66	(7)
	HDD10.0	4	0	0	0	0	0	1	2	1	0	0	0	0	(8)
	HDD18.3	832	2	1	5	37	94	153	175	163	114	62	24	3	(9)
	CDD10.0	2539	348	318	309	224	166	98	86	96	137	200	241	317	(10)
	CDD18.3	326	92	85	56	11	2	0	0	0	1	3	14	62	(11)
	CDH23.3	290	114	73	34	4	2	0	1	0	3	4	2	53	(12)
	CDH26.7	7	1	1	2	1	0	0	0	0	0	1	0	2	(13)
(14) Wind	WSAvg	4.7	5.1	4.9	4.9	4.5	4.2	4.0	4.2	4.3	4.7	5.1	5.2	5.0	(14)
(15) Precipitation	PrecAvg	450	24	18	28	49	51	56	61	48	31	37	28	19	(15)
	PrecMax	588	126	64	136	255	146	140	140	130	70	111	92	103	(16)
	PrecMin	262	0	2	4	9	12	17	20	14	8	5	4	2	(17)
	PrecStd	84	30	12	31	49	26	29	30	25	16	30	23	21	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.5	25.8	25.3	23.4	22.2	20.8	20.5	20.3	21.9	22.7	23.2	25.9	(19)
		MCWB	22.0	20.9	20.9	18.1	16.0	14.8	14.8	14.3	15.8	16.7	18.6	20.4	(20)
	2%	DB	25.3	24.8	24.2	21.8	20.5	18.7	18.5	18.3	19.4	20.9	22.2	24.6	(21)
		MCWB	21.0	20.6	20.4	18.4	16.2	14.3	14.3	14.0	15.3	16.9	18.1	20.0	(22)
	5%	DB	24.5	24.2	23.4	20.9	19.5	17.5	17.2	17.2	18.3	20.0	21.4	23.8	(23)
		MCWB	20.4	20.3	20.0	18.1	16.1	14.2	13.9	13.7	14.9	16.5	17.5	19.4	(24)
	10%	DB	23.7	23.6	22.6	20.1	18.5	16.6	16.2	16.2	17.5	19.1	20.7	22.9	(25)
		MCWB	19.9	19.9	19.5	17.7	15.8	13.9	13.5	13.4	14.5	16.0	17.0	18.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.1	22.6	22.3	20.7	18.7	16.2	15.7	16.1	17.7	19.0	21.1	22.2	(27)
		MCDB	25.5	24.4	23.6	21.3	19.7	18.0	17.8	17.9	20.0	21.7	22.2	24.0	(28)
	2%	WB	22.2	21.8	21.5	19.7	17.6	15.6	15.1	15.3	16.7	18.1	19.4	21.2	(29)
		MCDB	24.5	23.7	23.2	20.8	19.0	17.3	17.3	17.0	18.8	20.1	21.3	23.4	(30)
	5%	WB	21.5	21.3	20.9	19.0	17.0	15.1	14.5	14.6	15.9	17.4	18.4	20.5	(31)
		MCDB	23.8	23.3	22.7	20.3	18.6	16.8	16.7	16.4	17.8	19.3	20.5	22.9	(32)
	10%	WB	20.7	20.8	20.2	18.4	16.4	14.4	14.0	14.0	15.1	16.8	17.7	19.9	(33)
		MCDB	23.1	23.0	22.0	19.9	18.2	16.3	16.1	16.0	17.2	18.7	20.0	22.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	4.4	4.7	4.9	6.0	7.3	7.1	6.7	5.6	4.8	4.8	4.9	(35)
		MCDBR	5.3	5.1	5.8	5.8	7.7	8.9	8.6	8.1	6.7	5.7	5.6	5.8	(36)
	5% WB	MCWBR	3.0	2.9	3.3	3.5	4.4	5.5	5.6	4.9	3.9	3.2	3.1	3.2	(37)
		MCWBR	4.7	4.1	4.2	4.3	5.6	7.0	7.6	6.4	6.0	4.6	4.4	4.8	(38)
(40) Clear-Sky Solar Irradiance	taub		0.362	0.365	0.353	0.335	0.328	0.321	0.321	0.340	0.352	0.369	0.355	0.362	(40)
		taud	2.507	2.508	2.539	2.574	2.574	2.569	2.560	2.499	2.446	2.421	2.494	2.496	(41)
	Ebn at Noon		974	949	924	885	834	813	833	864	909	934	974	978	(42)
		Edh at Noon	113	109	98	84	74	70	74	89	106	117	114	115	(43)
(44) All-Sky Solar Radiation	RadAvg		7.50	6.68	5.38	3.99	2.90	2.46	2.70	3.49	4.78	6.02	7.18	7.73	(44)
	RadStd		0.26	0.26	0.28	0.18	0.15	0.13	0.12	0.16	0.22	0.34	0.28	0.34	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	+0.36	+0.49	N/A	N/A	+0.53	+0.47	-2	-74	+111	+61

Nomenclature: See separate page